

Operator-Theoretic Continuation on the Alaoglu–Erdős Conjecture

Local universality, arithmetic Hamiltonians, affine rigidity,
Jordan self-extensions, and Koopman realizations

Research continuation report

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Abstract

Let $x \in \mathbb{R}$ and suppose that $A = 2^x$ and $B = 3^x$ are integers. The Alaoglu–Erdős conjecture asserts that x is an integer. This document is a continuation of the supplied unified report, not a replacement for it. It deliberately avoids repeating that report’s historical survey, standard transcendence-theoretic reductions, extensive catalogue of equivalent formulations, and previously explored analytic, p -adic, matrix, and asymptotic routes.

The new organizing question is whether the scalar data can be forced to extend to an operator structure carrying enough *additive*, *infinitesimal*, or *finite-lattice* coherence to make a nonintegral exponent impossible. Several exact results are obtained. Pairs of bilateral multiplication shifts are universal and therefore spectrally blind to the arithmetic labels. The two-mode Fock Hamiltonians for $(2, 3)$ and (A, B) are exactly related by time rescaling, so heat traces, local KMS data, and Weyl asymptotics supply no obstruction. In contrast, a global isometric intertwiner for the arithmetic Hamiltonian $He_n = (\log n)e_n$ exists precisely for positive integral scaling factors. A finite prime-mode completion makes x an eigenvalue of an integer valuation matrix and hence an integer; an automorphic completion forces $x = 1$. Adjoining even one affine $ax + b$ coherence relation is quantitatively rigid, including modulo compact operators.

Finite-dimensional functional calculus exhibits a complementary dichotomy. Every semisimple rational operator supported at the two spectral points 2 and 3 is automatically blind: its x -th power is rational whenever 2^x and 3^x are. A single nontrivial Jordan self-extension probes the derivative $x2^{x-1}$ or $x3^{x-1}$, and rationality of that first jet forces $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$. For the fractional part $\alpha = \{x\}$, two rational-valued weighted Koopman operators are constructed. Their cocycles are bounded coboundaries, explaining why spectra, Lyapunov growth, entropy, and determinant-like invariants collapse. The surviving datum is a projection of trace α : an exact finite-dimensional tracial model or an exact finite-Hankel realization would force α rational, whereas ordinary matrix approximation would not.

No unconditional proof of the conjecture is claimed. The report instead isolates a sharp local-to-global obstruction and ranks concrete closing targets. The strongest small targets are: propagation to one third multiplicatively independent algebraic base; one rational Jordan self-extension; one finite prime-mode completion; one exact finite-Hankel realization; or one additive affine coherence relation. Each target is proved sufficient, and the seductive operator invariants that merely repackage the original 2×2 exponential array are explicitly separated from those that add genuinely new arithmetic information.

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1 Scope, conventions, and relation to the companion report

The supplied unified report already develops the classical reduction to a special case of the Four Exponentials Conjecture, the Gelfond–Schneider and Six Exponentials thresholds, logarithmic and valuation formulations, and a wide range of analytic and algebraic approaches. Repeating those discussions would obscure the purpose of a continuation. Here they are used only as black-box inputs when an operator construction reaches one of their exact closing thresholds.

The main novelty is the distinction between three levels of structure.

1. **Local multiplicative structure.** One retains only the two multiplicatively independent modes generated by 2 and 3. At this level, almost all natural operator invariants are universal or are changed only by a continuous rescaling of time.
2. **Finite-mode arithmetic structure.** One asks whether the two modes extend to a finite lattice of prime modes, to a nonsemisimple rational functional calculus, or to a finite linear realization. Any such exact extension has discrete algebraic invariants and is already strong enough to close the conjecture.
3. **Additive–multiplicative structure.** One couples multiplication by n to translation by 1, as in the affine semigroup. Integer labels then become rigid rather than merely spectral parameters.

All Hilbert spaces are complex and separable. For an unbounded operator, an intertwining identity is initially interpreted on the algebraic span of the indicated standard basis, which is a common invariant core. The positive integers are $\mathbb{N} = \{1, 2, \dots\}$, while $\mathbb{N}_0 = \{0, 1, \dots\}$.

1.1 Minimal standing reduction

Set

$$A = 2^x \in \mathbb{N}, \quad B = 3^x \in \mathbb{N}.$$

If $x < 0$, then $0 < 2^x < 1$, impossible for an integer; hence $x \geq 0$. The case $x = 0$ is already integral, so the substantive case has $x > 0$. Write

$$x = n + \alpha, \quad n = \lfloor x \rfloor, \quad 0 \leq \alpha < 1.$$

Then

$$r := 2^\alpha = \frac{A}{2^n} \in \mathbb{Q}, \quad s := 3^\alpha = \frac{B}{3^n} \in \mathbb{Q}. \tag{1}$$

Lemma 1.1 (Rational-exponent collapse). *If $b \geq 2$ is an integer, $y \in \mathbb{Q}$, and $b^y \in \mathbb{Q}$, then the denominator of y divides every valuation of b . In particular, if b is not a perfect power and $b^y \in \mathbb{Q}$, then $y \in \mathbb{Z}$. Consequently, under (1), $\alpha \in \mathbb{Q}$ implies $\alpha = 0$.*

Proof. Write $y = p/q$ in lowest terms with $q > 0$, and write $b^y = u/v$ in lowest terms. Taking q -th powers and applying each prime valuation gives

$$q(v_\ell(u) - v_\ell(v)) = p v_\ell(b).$$

Since $\gcd(p, q) = 1$, the integer q divides every $v_\ell(b)$. For $b = 2$, this gives $q = 1$. Since $0 \leq \alpha < 1$, one then has $\alpha = 0$. $\square$

Thus a counterexample would have x and α irrational; by Gelfond–Schneider it would in fact have x transcendental [10, 21, 2]. The point of the operator constructions below is not to rediscover this reduction, but to identify what additional exact structure would force the missing rationality.

1.2 A result ledger

Structure	Exact conclusion	Role
Bilateral shifts for two independent labels	All such pairs are unitarily equivalent with countable multiplicity.	No-go: pure spectrum is blind.
Two-mode Fock Hamiltonian	$H_{A,B} = xH_{2,3}$; heat traces and local KMS data are time-rescaled.	No-go: thermodynamics is blind.
Arithmetic Hamiltonian on all $\mathbb{N}$	An isometry T with $HT = xTH$ exists exactly for $x \in \mathbb{N}$.	Global closing criterion.
Finite invariant prime lattice	x is an algebraic-integer eigenvalue of a valuation matrix, hence $x \in \mathbb{Z}$.	Finite-mode closing criterion.
Unimodular prime-lattice completion	The same eigenvalue is an algebraic unit; hence $x = 1$.	Automorphic closing criterion.
Affine $ax + b$ coherence	Preserving $V_2S = S^2V_2$, even modulo compacts in the stated model, forces $A = 2$, hence $x = 1$.	Robust additive rigidity.
Semisimple rational functional calculus	Every operator with minimal polynomial dividing $(t - 2)(t - 3)$ remains rational under $t \mapsto t^x$.	No-go: two values suffice.
Jordan self-extension	Rationality of $(2I + N)^x$, $N^2 = 0 \neq N$, forces $x \in \mathbb{Q}$.	Infinitesimal closing criterion.
Sturmian Koopman realization	The rational weights are bounded coboundaries; one projection has trace α .	Separates blind invariants from trace.
Exact finite tracial or Hankel model	The trace/frequency α is rational, so $\alpha = 0$.	Finite-realization closing criterion.
One third independent algebraic base	Six Exponentials forces $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$.	Smallest high-value propagation target.

2 The maximal two-mode algebraic sector

Before introducing operator models, it is useful to sharpen exactly how far the hypotheses propagate by known transcendence theory. This result is the correct boundary condition for every later local-to-global construction.

Theorem 2.1 (Three-mode barrier). *Assume $x \notin \mathbb{Q}$, $2^x, 3^x \in \overline{\mathbb{Q}}$, and let $q \in \mathbb{Q}_{>0}$. Then*

$$q^x \in \overline{\mathbb{Q}} \iff q = 2^m 3^n \text{ for some } m, n \in \mathbb{Z}.$$

In particular, if a nonintegral solution of the Alaoglu–Erdős hypotheses exists, then

$$\{k \in \mathbb{N} : k^x \in \overline{\mathbb{Q}}\} = \{2^m 3^n : m, n \in \mathbb{N}_0\}.$$

Every positive integer containing a prime other than 2 or 3 has transcendental x -th power.

Proof. The reverse implication is immediate: $(2^m 3^n)^x = (2^x)^m (3^x)^n$ is algebraic.

For the forward implication, suppose that q is multiplicatively independent of 2 and 3. Then $\log 2, \log 3, \log q$ are linearly independent over $\mathbb{Q}$, while $1, x$ are linearly independent over $\mathbb{Q}$. The Six Exponentials Theorem [22, Chap. 3] says that at least one entry of

$$\begin{pmatrix} \exp(\log 2) & \exp(\log 3) & \exp(\log q) \\ \exp(x \log 2) & \exp(x \log 3) & \exp(x \log q) \end{pmatrix} = \begin{pmatrix} 2 & 3 & q \\ 2^x & 3^x & q^x \end{pmatrix}$$

is transcendental, contradicting the assumed algebraicity of all six entries. Therefore q is multiplicatively dependent with 2, 3: there are integers a, b, c , with $c \neq 0$, such that $q^c 2^a 3^b = 1$. Prime valuations show that q has no prime factor other than 2 or 3; since $q \in \mathbb{Q}_{>0}$, it follows that $q = 2^m 3^n$ with $m, n \in \mathbb{Z}$. $\square$

Remark 2.2. The theorem says more than “a third prime would help.” Under a hypothetical counterexample, the rank-two group $\langle 2, 3 \rangle$ is the entire rational domain on which the x -power map remains algebraic. An operator proof must therefore force leakage out of this maximal sector, or extract an infinitesimal or finite-dimensional invariant that cannot live inside it.

Corollary 2.3 (One-vector closing target). *Under the Alaoglu–Erdős hypotheses, it suffices to prove $q^x \in \overline{\mathbb{Q}}$ for one rational $q > 0$ having a prime factor other than 2 or 3. In particular, any one of $5^x, 7^x, 10^x, \dots$ being algebraic closes the conjecture.*

3 Bilateral multiplication shifts: a universality theorem

Let $G = \mathbb{Q}_{>0}$ with multiplication, and let $\mathcal{H}_G = \ell^2(G)$ with standard basis $(e_q)_{q \in G}$. For $a \in G$, define a unitary

$$U_a e_q = e_{aq}.$$

These are the most literal bilateral operator representatives of multiplication by rational labels; the orbit decomposition below is the regular-representation viewpoint standard in semigroup operator theory [18, 14].

Theorem 3.1 (Universality of two independent multiplication shifts). *If $a, b \in \mathbb{Q}_{>0}$ are multiplicatively independent, then the commuting pair (U_a, U_b) is unitarily equivalent to a countably infinite multiple of the left regular representation of $\mathbb{Z}^2$. Consequently, for any two multiplicatively independent pairs (a, b) and (c, d) , the pairs (U_a, U_b) and (U_c, U_d) are unitarily equivalent.*

Proof. Let $\Lambda = \langle a, b \rangle = \{a^m b^n : m, n \in \mathbb{Z}\}$. Multiplicative independence identifies Λ with $\mathbb{Z}^2$. Choose a transversal C for the cosets G/Λ . Then

$$\ell^2(G) = \bigoplus_{c \in C} \ell^2(c\Lambda),$$

and on each summand the unitary

$$e_{ca^m b^n} \mapsto e_{(m,n)}$$

conjugates U_a, U_b to the two coordinate bilateral shifts on $\ell^2(\mathbb{Z}^2)$. The quotient G/Λ is countably infinite: it is countable, and primes outside the finite prime support of ab give infinitely many distinct cosets. Thus the multiplicity is countably infinite and does not depend on the labels. $\square$

For $x > 0$, the integers $A = 2^x$ and $B = 3^x$ are multiplicatively independent. Indeed, $A^m B^n = 1$ implies $(2^m 3^n)^x = 1$, hence $m = n = 0$. Therefore [Theorem 3.1](#) applies equally to $(2, 3)$ and (A, B) .

Corollary 3.2 (Spectral blindness). *The joint spectrum, joint spectral type, von Neumann algebra generated by the pair, and spectral multiplicity of the bilateral pair do not distinguish $(2, 3)$ from $(2^x, 3^x)$. Any proposed proof based only on the unitary-equivalence class of the two multiplication shifts cannot see whether x is integral.*

Remark 3.3. Under Fourier transform, one copy of the regular representation becomes multiplication by the two coordinate functions on $L^2(\mathbb{T}^2)$. Thus the joint spectrum is always $\mathbb{T}^2$, with the same Haar spectral measure and the same countable multiplicity. The failure is not a lack of a sufficiently clever spectral invariant; the entire pair is universal.

3.1 The logarithmic Hamiltonian restores scale locally

Define the self-adjoint arithmetic Hamiltonian

$$H_G e_q = (\log q) e_q.$$

On its natural core,

$$[H_G, U_a] = (\log a) U_a. \quad (2)$$

The commutator remembers the logarithmic label that pure shift spectrum forgets. Nevertheless, on the two generated orbits it still gives no obstruction.

Let

$$\mathcal{H}_{2,3} = \overline{\text{span}}\{e_{2^m 3^n} : m, n \in \mathbb{Z}\}, \quad \mathcal{H}_{A,B} = \overline{\text{span}}\{e_{A^m B^n} : m, n \in \mathbb{Z}\}.$$

The map

$$W e_{2^m 3^n} = e_{A^m B^n} \quad (3)$$

is a unitary $\mathcal{H}_{2,3} \rightarrow \mathcal{H}_{A,B}$, and

$$H_G W = x W H_G. \quad (4)$$

Thus even the Hamiltonian-covariant equivalence is automatic on the maximal rank-two algebraic sector. The arithmetic content lies in extending (3) beyond that sector while preserving a discrete basis or lattice.

4 Two-mode Fock Hamiltonians and thermodynamic no-go results

Let $\mathcal{F}_2 = \ell^2(\mathbb{N}_0^2)$, with number operators

$$N_1 e_{m,n} = m e_{m,n}, \quad N_2 e_{m,n} = n e_{m,n}.$$

For multiplicatively independent $a, b > 1$, set

$$H_{a,b} = (\log a)N_1 + (\log b)N_2.$$

Its eigenvalue at $e_{m,n}$ is $\log(a^m b^n)$, so the spectrum is simple.

Proposition 4.1 (Exact time homothety). *Under $A = 2^x$ and $B = 3^x$,*

$$H_{A,B} = xH_{2,3}$$

as self-adjoint operators on the same Fock space.

This equality is stronger than an asymptotic similarity. Every spectral quantity derived functorially from H is merely reparametrized.

For $\beta > 0$, the heat trace is

$$Z_{a,b}(\beta) := \text{Tr}(e^{-\beta H_{a,b}}) = \sum_{m,n \geq 0} a^{-\beta m} b^{-\beta n} = \frac{1}{(1 - a^{-\beta})(1 - b^{-\beta})}. \quad (5)$$

Consequently,

$$Z_{A,B}(\beta) = Z_{2,3}(x\beta).$$

As $\beta \downarrow 0$,

$$Z_{a,b}(\beta) = \frac{1}{\beta^2 \log a \log b} + O(\beta^{-1}), \quad (6)$$

and the eigenvalue-counting function satisfies the elementary lattice estimate

$$N_{a,b}(E) := \#\{(m, n) \in \mathbb{N}_0^2 : m \log a + n \log b \leq E\} = \frac{E^2}{2 \log a \log b} + O(E). \quad (7)$$

Both leading terms simply acquire the continuous factor x^{-2} when $(a, b) = (2, 3)$ is replaced by (A, B) .

Let L_1, L_2 be the coordinate unilateral shifts on $\mathcal{F}_2$. The Toeplitz algebra $\mathcal{T}^{(2)} = C^*(L_1, L_2)$ admits the dynamics

$$\sigma_t^{a,b}(L_1) = a^{it} L_1, \quad \sigma_t^{a,b}(L_2) = b^{it} L_2.$$

Then

$$\sigma_t^{A,B} = \sigma_{xt}^{2,3}.$$

Thus the underlying C^* -algebra, its K -theory, and the local two-mode representation are identical; the dynamics differs only by a time change. Gibbs and KMS data parameterized by inverse temperature are transported by $\beta \mapsto x\beta$; see [4, 3] for the general equilibrium formalism.

Warning 4.2 (Why a critical temperature cannot quantize x). A critical inverse temperature is meaningful only after fixing an external normalization of time or energy. In the isolated two-mode system there is no such normalization: multiplying the Hamiltonian by an arbitrary positive real number is an exact time reparametrization. Therefore a pole, heat coefficient, entropy density, or KMS transition inside this sector cannot force $x \in \mathbb{Z}$.

5 The full prime Fock space and the local-to-global gap

Let $\mathcal{P}$ be the set of primes and

$$\mathbb{N}_0^{(\mathcal{P})} = \{v = (v_p)_{p \in \mathcal{P}} : v_p \in \mathbb{N}_0, v_p = 0 \text{ for all but finitely many } p\}.$$

Unique factorization identifies $\mathbb{N}$ with this monoid through the valuation vector

$$\nu(n) = (v_p(n))_{p \in \mathcal{P}}.$$

The full prime Fock space is

$$\mathcal{F}_{\mathcal{P}} = \ell^2(\mathbb{N}_0^{(\mathcal{P})}) \cong \ell^2(\mathbb{N}),$$

and its Hamiltonian is

$$H_{\mathcal{P}} e_v = \left(\sum_{p \in \mathcal{P}} v_p \log p \right) e_v.$$

Under the identification with $\ell^2(\mathbb{N})$, this is simply $H e_n = (\log n) e_n$.

Let $a = \nu(A)$ and $b = \nu(B)$. Since A, B are multiplicatively independent, the map

$$J : \mathcal{F}_2 \longrightarrow \mathcal{F}_{\mathcal{P}}, \quad J e_{m,n} = e_{ma+nb}$$

is an isometry. Moreover,

$$H_{\mathcal{P}} J = x J H_{2,3}. \tag{8}$$

Thus the hypotheses themselves give an exact Hamiltonian-covariant embedding of the two-mode sector into the arithmetic spectrum. This is not a partial proof; it is the canonical operator form of the two scalar equalities.

The missing step is now precise:

Can the two-column valuation map $e_2 \mapsto \nu(A)$, $e_3 \mapsto \nu(B)$ be forced to extend to a finite or global integral lattice map that rescales the logarithmic energy functional by x ?

Abstract Hilbert-space extension theorems are irrelevant here. The isometry J can always be extended abstractly after adding arbitrary orthogonal complements. What matters is an extension that remains *monomial on the arithmetic basis, integral on valuation lattices, and covariant for $H_{\mathcal{P}}$* . The next two sections show that either a global extension or a finite prime-mode extension would close the conjecture.

6 Global arithmetic-Hamiltonian intertwiners

On $\ell^2(\mathbb{N})$, let

$$H e_n = (\log n) e_n,$$

with domain

$$\mathcal{D}(H) = \left\{ \xi : \sum_{n \geq 1} (\log n)^2 |\xi_n|^2 < \infty \right\}.$$

The eigenvalues are simple. This converts a Hamiltonian covariance identity into an exact arithmetic condition on every basis label.

Theorem 6.1 (Global isometric covariance). *For $c > 0$, the following are equivalent.*

1. *There is an isometry $T : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ such that $HT = cTH$ on the finite-support core.*
2. *$n^c \in \mathbb{N}$ for every $n \in \mathbb{N}$.*
3. *$c \in \mathbb{N}$.*

For $c = k \in \mathbb{N}$, a canonical intertwiner is

$$T_k e_n = e_{n^k}.$$

Proof. Suppose $HT = cTH$. For each n ,

$$H(Te_n) = c(\log n)Te_n.$$

Since T is an isometry, $Te_n \neq 0$. The spectrum of H is simple and consists of the numbers $\log m$, $m \in \mathbb{N}$, so there is an $m \in \mathbb{N}$ with $\log m = c \log n$; hence $m = n^c \in \mathbb{N}$. This proves (1) $\Rightarrow$ (2).

To prove (2) $\Rightarrow$ (3), suppose that $c \notin \mathbb{N}$, and choose $r = \lfloor c \rfloor + 1$. The r -th forward difference

$$\Delta^r n^c = \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} (n+j)^c$$

is an integer. Repeated use of the fundamental theorem of calculus gives the exact integral formula

$$\Delta^r n^c = c(c-1) \cdots (c-r+1) \int_{[0,1]^r} (n+t_1+\cdots+t_r)^{c-r} dt_1 \cdots dt_r.$$

The coefficient is nonzero because c is not an integer, and the integrand has fixed sign. Thus the difference is nonzero for every n , while $c-r < 0$ implies $\Delta^r n^c \rightarrow 0$. For large n this is a nonzero integer of absolute value less than 1, a contradiction. Hence $c \in \mathbb{N}$.

Finally, if $c = k \in \mathbb{N}$, then $n \mapsto n^k$ is injective, so T_k is an isometry, and

$$HT_k e_n = k(\log n)e_{n^k} = kT_k H e_n.$$

□

Remark 6.2 (The endpoint $c = 0$). The scalar map $n \mapsto n^0 = 1$ collapses all basis vectors and cannot be implemented by an isometry. It is represented by a rank-one collapse rather than an endomorphism. This is why [Theorem 6.1](#) classifies positive integers; the original conjecture separately includes the already trivial solution $x = 0$.

6.1 Partial intertwiners and exact support

For arbitrary $c > 0$, define

$$I_c = \{n \in \mathbb{N} : n^c \in \mathbb{N}\}.$$

There is a canonical isometry

$$T_c : \ell^2(I_c) \longrightarrow \ell^2(\mathbb{N}), \quad T_c e_n = e_{n^c},$$

and it satisfies $HT_c = cT_cH$ on the natural source core. Under the Alaoglu–Erdős hypotheses, $\{2^m 3^n : m, n \in \mathbb{N}_0\} \subseteq I_x$, so an infinite-rank Hamiltonian intertwiner already exists.

Combining this observation with [Theorem 2.1](#) gives a complete classification under a hypothetical counterexample.

Corollary 6.3 (Maximal support of the partial intertwiner). *If x is a nonintegral solution of the Alaoglu–Erdős hypotheses, then*

$$I_x = \{2^m 3^n : m, n \in \mathbb{N}_0\}.$$

More strongly, replacing “integer” by “algebraic” in the definition of I_x gives the same set.

This identifies an especially small operator target: it is enough to make the covariant partial intertwiner nonzero at *one* basis vector e_q with q not 3-smooth and with algebraic target label. No global extension is needed; [Theorem 2.3](#) then closes the conjecture.

7 Finite prime-mode closure and valuation matrices

The global criterion in [Theorem 6.1](#) is much stronger than the original hypotheses. A finite-dimensional lattice completion is already enough.

Let $P = \{p_1, \dots, p_d\}$ be a finite set of primes containing 2. Write

$$\Gamma_P = \left\{ \prod_{j=1}^d p_j^{v_j} : v_j \in \mathbb{Z} \right\} \cong \mathbb{Z}^d.$$

Theorem 7.1 (Finite prime closure). *Let $c \in \mathbb{R} \setminus \{0\}$. Suppose that*

1. $2 \in P$;
2. $p_j^c \in \Gamma_P$ for every $p_j \in P$; and
3. 2^c is algebraic (in particular, rational or integral).

Then $c \in \mathbb{Z}$. If $c \geq 0$, then $c \in \mathbb{N}_0$.

Proof. For each j , write uniquely

$$p_j^c = \prod_{i=1}^d p_i^{m_{ij}}, \quad m_{ij} \in \mathbb{Z},$$

and let $M = (m_{ij}) \in M_d(\mathbb{Z})$. Set

$$\ell = (\log p_1, \dots, \log p_d)^\top.$$

Taking real logarithms column by column gives

$$M^\top \ell = c\ell.$$

Thus c is an eigenvalue of an integer matrix and therefore an algebraic integer. If c were algebraic irrational, Gelfond–Schneider would make 2^c transcendental, contrary to assumption. Hence $c \in \mathbb{Q}$. A rational algebraic integer is an integer. $\square$

Corollary 7.2 (Automorphic closure). *In the setting of Theorem 7.1, suppose additionally that the power map*

$$\Gamma_P \longrightarrow \Gamma_P, \quad q \longmapsto q^c,$$

is onto. Then $M \in \mathrm{GL}_d(\mathbb{Z})$, so c is an algebraic unit. If $c > 0$, then $c = 1$.

Proof. Surjectivity makes the valuation matrix unimodular. Every eigenvalue of a unimodular integer matrix is an algebraic unit. By Theorem 7.1, c is a rational integer; the only positive rational integer that is an algebraic unit is 1. $\square$

7.1 Operator form of the valuation theorem

On $\ell^2(\mathbb{Z}^d)$, define

$$H_\ell e_v = \langle \ell, v \rangle e_v.$$

The power map on Γ_P is injective for $c \neq 0$, hence the integer matrix M is injective on $\mathbb{Z}^d$. Therefore

$$T_M e_v = e_{Mv}$$

is an isometry, and

$$H_\ell T_M = c T_M H_\ell \iff M^\top \ell = c \ell.$$

Thus the theorem may be read as follows: a finite arithmetic lattice model of Hamiltonian time rescaling makes the scaling factor an algebraic integer. The scalar algebraicity $2^c \in \overline{\mathbb{Q}}$ then reduces it to an ordinary integer.

If all p_j^c are integers supported on P , then M has nonnegative entries and the same statement can be realized on the positive Fock cone $\ell^2(\mathbb{N}_0^d)$. If the matrix is irreducible, the positive logarithm vector exhibits c as its Perron–Frobenius eigenvalue. Perron–Frobenius theory does not itself prove integrality; the decisive extra input remains Gelfond–Schneider.

Closing criterion 7.3 (Finite propagation closure). *Starting from $P_0 = \{2, 3\}$, suppose one can prove that p^x is rational for every prime introduced as a divisor of a previously known q^x , and that the resulting prime-propagation process terminates in a finite set P . Then P is invariant under the x -power map, so Theorem 7.1 proves $x \in \mathbb{Z}$.*

A counterexample therefore cannot admit any finite exact prime-substitution completion. This gives a concrete diagnostic for operator correspondences, graph C^* -algebras, fusion matrices, and substitution systems: if their finite incidence matrix really encodes the logarithmic scaling, the conjecture is already over.

8 C^* -dynamical interpretation: local covariance versus global arithmetic

The preceding Hilbert-space constructions have a direct C^* -dynamical reading. It is useful to state it without relying on a particular presentation of the Bost–Connes algebra.

Let $\mathfrak{T}_{\mathrm{mult}}$ be the concrete Toeplitz algebra on $\ell^2(\mathbb{N})$ generated by the multiplication isometries

$$V_m e_n = e_{mn}, \quad m, n \in \mathbb{N}.$$

They satisfy $V_m V_n = V_{mn}$. The Hamiltonian $H e_n = (\log n) e_n$ implements the time evolution

$$\sigma_t(V_m) = m^{it} V_m. \quad (9)$$

On the algebraic, nonselfadjoint semigroup generated by V_2, V_3 , the assignment

$$V_2 \mapsto V_A, \quad V_3 \mapsto V_B$$

respects products and satisfies

$$\sigma_t(\Phi(V_j)) = \Phi(\sigma_{xt}(V_j)), \quad j = 2, 3.$$

At the C^* -level one should use the two-coordinate Fock compression from [Theorem 4.1](#): adjoints and range projections in the full concrete representation can additionally record gcd/lcm data. That extra data is precisely valuation-lattice data; it does not follow from abstract rank-two dynamics alone. In either formulation, local time covariance merely rescales the two known energies and does not quantize x .

A global monomial endomorphism would have to send each V_n to V_{n^x} . Such labels exist for all n exactly when $n^x \in \mathbb{N}$ for all n , and [Theorem 6.1](#) then forces $x \in \mathbb{N}$. Likewise, a finite prime-generated endomorphism gives the integer matrix of [Theorem 7.1](#).

Remark 8.1 (Bost–Connes and related systems). In Bost–Connes-type systems [\[3\]](#), the multiplicative isometries carry the same logarithmic time evolution, while additional additive or torsion data encode genuinely arithmetic relations. Restricting to the two multiplicative modes loses precisely that extra coherence and leaves only a time-rescaled rank-two system. A useful operator strategy must therefore exploit the additive diagonal, transfer operators, or an integral lattice completion; KMS data from the isolated modes cannot do the job.

9 Affine $ax + b$ rigidity

The cleanest demonstration that additive structure restores integer labels uses the standard affine representation familiar from $ax + b$ semigroup C^* -algebras [\[8, 6\]](#) on $\ell^2(\mathbb{N}_0)$. Define

$$S e_k = e_{k+1}, \quad V_m e_k = e_{mk}.$$

Then

$$V_m S = S^m V_m. \quad (10)$$

Unlike the purely multiplicative relations, this relation records the integer m as an actual translation length.

Theorem 9.1 (Quantitative affine rigidity). *For integers $a, b, d \geq 1$, set*

$$D_{a,b,d} = V_a S^d - S^{bd} V_a.$$

Then

$$D_{a,b,d} = 0 \iff a = b.$$

If $a \neq b$, then $D_{a,b,d}$ is not compact and

$$\|D_{a,b,d}\|_{\text{ess}} \geq \sqrt{2}.$$

In particular, the affine relation determines the multiplicative label even in the Calkin algebra and is stable under norm errors smaller than $\sqrt{2}$.

Proof. For every $k \geq 0$,

$$D_{a,b,d}e_k = e_{a(k+d)} - e_{ak+bd} = e_{ak+ad} - e_{ak+bd}.$$

If $a = b$, this is zero. If $a \neq b$, the two displayed basis vectors are distinct, so

$$\|D_{a,b,d}e_k\| = \sqrt{2}$$

for every k . Since $e_k \rightarrow 0$ weakly and every compact operator maps this sequence to a norm-null sequence, for every compact K ,

$$\liminf_{k \rightarrow \infty} \|(D_{a,b,d} - K)e_k\| \geq \sqrt{2}.$$

Taking the infimum over K proves the essential-norm bound and noncompactness. $\square$

Corollary 9.2 (One affine coherence closes the conjecture). *Suppose a proposed operator extension sends*

$$S \mapsto S^d, \quad V_2 \mapsto V_A$$

for some $d \geq 1$, and preserves the single relation $V_2S = S^2V_2$, either exactly or modulo compact operators. Then $A = 2$, hence $x = 1$.

Proof. The image of the relation requires $V_AS^d - S^{2d}V_A$ to vanish, or at least to be compact. Apply [Theorem 9.1](#) with $a = A$ and $b = 2$. $\square$

Remark 9.3. This criterion is intentionally much stronger than the original scalar hypotheses, but it is structurally minimal: one generator S , one multiplication mode V_2 , and one coherence square suffice. It also explains why a proof cannot emerge from the multiplicative semigroup alone. Translation gives an external integral ruler against which the label A can be measured.

9.1 A categorical shadow

Let E_n denote an n -dimensional object in a semisimple tensor category with direct sums. The elementary coherence

$$E_2 \otimes E_2 \cong E_2 \oplus E_2$$

expresses $2 \cdot 2 = 2 + 2$. If a dimension-scaling construction sends E_2 to an object of dimension $A = 2^x$ and preserves this one isomorphism, then

$$A^2 = 2A.$$

Since $A > 0$, one gets $A = 2$ and $x = 1$. The statement is simple, but its lesson is exact: a single additive–multiplicative coherence is vastly more rigid than all multiplicative powers of the two generators.

10 Finite-dimensional functional calculus: semisimple blindness

A direct matrix formulation replaces the two scalars by $\text{diag}(2, 3)$ and observes that $\text{diag}(2, 3)^x = \text{diag}(A, B)$. By itself this is exactly the original problem. More revealing is the behavior of all rational operators whose spectrum is supported at 2 and 3.

Let $f_x(t) = t^x$ on the positive real axis, extended holomorphically to the slit plane; standard matrix-function facts used below may be found in [11] $\mathbb{C} \setminus (-\infty, 0]$.

Theorem 10.1 (Semisimple two-point calculus is automatic). *Assume $A = 2^x \in \mathbb{Q}$ and $B = 3^x \in \mathbb{Q}$. If $T \in M_d(\mathbb{Q})$ satisfies*

$$(T - 2I)(T - 3I) = 0,$$

then

$$T^x = (B - A)T + (3A - 2B)I \in M_d(\mathbb{Q}). \quad (11)$$

If T, A, B have integer entries, then T^x has integer entries.

Proof. The minimal polynomial is square-free, so T is semisimple with spectral projections

$$P_2 = 3I - T, \quad P_3 = T - 2I.$$

Since $P_2 + P_3 = I$,

$$T^x = AP_2 + BP_3 = A(3I - T) + B(T - 2I),$$

which is (11). □

The theorem covers every rationally diagonalizable coupling of the two modes, not merely the diagonal matrix. For example,

$$T_c = \begin{pmatrix} 2 & c \\ 0 & 3 \end{pmatrix} \quad (c \in \mathbb{Z})$$

is diagonalizable and

$$T_c^x = \begin{pmatrix} A & c(B - A) \\ 0 & B \end{pmatrix}.$$

Thus adding an off-diagonal interaction between the distinct eigenvalues does not add arithmetic content. It merely evaluates the divided difference

$$f_x[2, 3] = \frac{3^x - 2^x}{3 - 2} = B - A,$$

which the original two scalar values already determine.

Remark 10.2 (Module-theoretic interpretation). Finite-dimensional $\mathbb{Q}[t]$ -modules supported at the distinct maximal ideals $(t - 2)$ and $(t - 3)$ split because the ideals are coprime. Functional calculus on a semisimple module depends only on the two values $f_x(2)$ and $f_x(3)$. No extension class remains on which a derivative could be detected.

11 Jordan self-extensions and the first-jet obstruction

A repeated spectral point changes the situation. Let $N \neq 0$ be a rational matrix with $N^2 = 0$. The binomial series terminates:

$$(bI + N)^x = b^x I + xb^{x-1}N. \quad (12)$$

Theorem 11.1 (One rational Jordan lift closes the conjecture). *Assume $b \geq 2$, $b^x \in \mathbb{Q}$, and $N \in M_d(\mathbb{Q})$ satisfies $N^2 = 0 \neq N$. If $(bI + N)^x \in M_d(\mathbb{Q})$, then $x \in \mathbb{Q}$. Consequently, under the Alaoglu–Erdős hypotheses, rationality of either $(2I + N)^x$ or $(3I + N)^x$ forces $x \in \mathbb{Z}$.*

Proof. Choose an entry $N_{ij} \neq 0$. From (12), rationality of the corresponding entry of $(bI + N)^x$ gives $xb^{x-1}N_{ij} \in \mathbb{Q}$, hence $xb^{x-1} \in \mathbb{Q}$. Since $b^x \in \mathbb{Q}^\times$,

$$x = \frac{b}{b^x} xb^{x-1} \in \mathbb{Q}.$$

Under the Alaoglu–Erdős hypotheses, Theorem 1.1 then gives $x \in \mathbb{Z}$. $\square$

Closing criterion 11.2 (Minimal infinitesimal bridge). *It is enough to prove that the scalar rationality $2^x \in \mathbb{Q}$ extends to the first-order dual-number identity*

$$(2 + \varepsilon)^x = 2^x + x2^{x-1}\varepsilon \in \mathbb{Q}[\varepsilon]/(\varepsilon^2).$$

Equivalently, it is enough to rationalize one nonzero tangent direction at the spectral point 2 (or at 3).

For a Jordan block of length r ,

$$(bI + N)^x = \sum_{j=0}^{r-1} \binom{x}{j} b^{x-j} N^j.$$

The first superdiagonal already reveals xb^{x-1} ; higher jets add no logical strength for rationality, although their divisibility may be useful in attempts to derive the first jet from an integral model.

11.1 The Loewner matrix diagnosis

Let $D = \text{diag}(2, 3)$. The Fréchet derivative of f_x at D is entrywise multiplication by the Loewner matrix:

$$(Df_x(D)[E])_{ij} = f_x^{[1]}(\lambda_i, \lambda_j) E_{ij}, \quad (\lambda_1, \lambda_2) = (2, 3),$$

where

$$f_x^{[1]}(\lambda, \mu) = \begin{cases} \frac{\lambda^x - \mu^x}{\lambda - \mu}, & \lambda \neq \mu, \\ x\lambda^{x-1}, & \lambda = \mu. \end{cases}$$

Under the scalar hypotheses, the off-diagonal coefficients are

$$f_x^{[1]}(2, 3) = f_x^{[1]}(3, 2) = B - A \in \mathbb{Z},$$

whereas the diagonal coefficients are

$$x2^{x-1}, \quad x3^{x-1}.$$

This is a precise separation:

Coupling the 2-mode to the 3-mode probes only a secant, already fixed by A and B . A self-coupling at either mode probes a derivative and immediately forces x rational if that derivative is rational.

In extension-theoretic language, extensions between the two distinct simple spectral modules split, while a self-extension is nontrivial and carries the missing first-jet datum.

11.2 A useful negative theorem

Proposition 11.3 (No semisimple matrix amplification can help). *Fix any finite family of rational matrices T_j whose minimal polynomials are square-free and whose spectra are subsets of $\{2, 3\}$. Under $2^x, 3^x \in \mathbb{Q}$, every T_j^x is rational. Therefore no condition asserting rationality or integrality of such semisimple powers is stronger than the original hypotheses.*

This rules out a broad class of seemingly richer finite-dimensional constructions. Any matrix-functional-calculus approach must introduce a repeated node, a new algebraic spectral point, or a nonfunctional relation such as addition.

12 The fractional-part system and rational Koopman cocycles

Return to $x = n + \alpha$, $0 \leq \alpha < 1$, and the rationals $r = 2^\alpha$, $s = 3^\alpha$ from (1). If $\alpha = 0$ the conjecture is proved, so assume for analysis that α is irrational.

Let $X = [0, 1)$ with Lebesgue measure, and let

$$R_\alpha(t) = \{t + \alpha\}.$$

Define the jump function

$$w(t) = \lfloor t + \alpha \rfloor = \mathbf{1}_{[1-\alpha, 1)}(t).$$

The elementary identity

$$w(t) - \alpha = t - R_\alpha(t) \tag{13}$$

is the source of both the usefulness and the limitations of the model.

For $b = 2, 3$, define

$$c_b(t) = b^{w(t) - \alpha}. \tag{14}$$

It takes exactly two values:

$$c_2(t) \in \left\{ \frac{1}{r}, \frac{2}{r} \right\} \subset \mathbb{Q}, \quad c_3(t) \in \left\{ \frac{1}{s}, \frac{3}{s} \right\} \subset \mathbb{Q}.$$

Conversely, rationality of either pair of values is equivalent to rationality of b^α .

Let U_α be the Koopman unitary

$$(U_\alpha \xi)(t) = \xi(R_\alpha(t))$$

on $L^2(X)$, and let

$$W_b = M_{c_b} U_\alpha.$$

Theorem 12.1 (The weighted operators are bounded coboundary twists). *Let $h_b(t) = b^t$. Then h_b and h_b^{-1} are bounded, and*

$$W_b = M_{h_b} U_\alpha M_{h_b}^{-1}. \tag{15}$$

Consequently W_b is similar to U_α , and all spectral invariants preserved by bounded similarity are independent of the rationality of b^α .

Proof. By (13),

$$c_b(t) = b^{t-R_\alpha(t)} = \frac{h_b(t)}{h_b(R_\alpha(t))}.$$

Substitution into the definition of W_b gives (15). $\square$

For irrational α , U_α has eigenvalues $e^{2\pi i k \alpha}$, $k \in \mathbb{Z}$, dense in $\mathbb{T}$, so

$$\text{Spec}(W_b) = \text{Spec}(U_\alpha) = \mathbb{T}.$$

The iterated products telescope:

$$\prod_{j=0}^{N-1} c_b(R_\alpha^j t) = b^{t-R_\alpha^N t}, \quad (16)$$

which is uniformly bounded above and below independently of N . Hence every logarithmic growth exponent is zero. Also

$$\int_X \log c_b dt = (\log b) \int_X (w - \alpha) dt = 0,$$

so the Fuglede–Kadison determinant of the positive multiplier M_{c_b} is 1. The rotation has Kolmogorov–Sinai entropy zero for every α [24]. None of these invariants can detect whether α is rational.

12.1 The common projection is the surviving datum

Let

$$P = M_w.$$

Then P is a projection and

$$\tau(P) = \int_X w(t) dt = \alpha. \quad (17)$$

The two rational multipliers are affine functions of the same projection:

$$M_{c_2} = \frac{1}{r}(I + P), \quad M_{c_3} = \frac{1}{s}(I + 2P). \quad (18)$$

Equivalently,

$$2rM_{c_2} - sM_{c_3} = I.$$

Thus the scalar hypotheses produce a rationally defined two-valued observable, but its invariant trace may be irrational. Rational operator coefficients do not imply rational values of a trace state.

Theorem 12.2 (Exact finite-dimensional trace criterion). *Assume (1). If there exist $N \geq 1$, a projection $P_N \in M_N(\mathbb{C})$, and a trace-preserving identification of the distinguished projection in the Koopman model such that*

$$\frac{1}{N} \text{Tr}(P_N) = \tau(P) = \alpha,$$

then $\alpha = 0$, and hence $x \in \mathbb{Z}$.

Proof. The normalized trace of a matrix projection is $\text{rank}(P_N)/N \in \mathbb{Q}$, so $\alpha \in \mathbb{Q}$. By Theorem 1.1 and $2^\alpha \in \mathbb{Q}$, $\alpha = 0$. $\square$

Warning 12.3 (Approximation is not enough). For every real $\alpha \in [0, 1]$, one can choose projections $P_N \in M_N(\mathbb{C})$ of rank $\lfloor \alpha N \rfloor$, giving

$$\frac{1}{N} \operatorname{Tr}(P_N) \longrightarrow \alpha.$$

Thus quasideagonal, sofic, MF, or moment approximation that reproduces the trace only in the limit cannot force rationality. An operator proof along this route needs an *exact* finite-dimensional trace realization, a bounded-denominator theorem, or an arithmetic restriction on the admissible matrix sizes.

13 Finite Hankel realization of the Sturmian jump word

Follow the orbit of 0 and define

$$w_k = w(R_\alpha^k 0) = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor \in \{0, 1\}. \quad (19)$$

This is the characteristic Sturmian word of slope α when α is irrational [17, 20]. It is recoverable from either rational cocycle in (14): the two values distinguish $w_k = 0$ from $w_k = 1$.

Let

$$\mathbf{H}_w = (w_{i+j})_{i,j \geq 0}$$

be its infinite Hankel matrix.

Theorem 13.1 (Finite Hankel rank closes the conjecture). *Under the Alaoglu–Erdős hypotheses, if $\operatorname{rank} \mathbf{H}_w < \infty$ over $\mathbb{C}$ (equivalently over any characteristic-zero field containing the sequence), then $\alpha = 0$, hence $x \in \mathbb{Z}$.*

Proof. Finite Hankel rank implies that (w_k) satisfies a linear recurrence of some finite order [12, 19] d , from some point onward. Once the recurrence is in force, the state

$$(w_k, w_{k+1}, \dots, w_{k+d-1})$$

belongs to the finite set $\{0, 1\}^d$ and determines the next term. Some state repeats, so the sequence is eventually periodic.

On the other hand, telescoping (19) gives

$$\frac{1}{N} \sum_{k=0}^{N-1} w_k = \frac{\lfloor N\alpha \rfloor}{N} \longrightarrow \alpha.$$

The limiting mean of an eventually periodic binary sequence is rational. Thus $\alpha \in \mathbb{Q}$, and Theorem 1.1 gives $\alpha = 0$. $\square$

Corollary 13.2 (No exact finite-dimensional linear realization). *If x were a counterexample, there could be no finite-dimensional linear system*

$$w_k = \lambda(T^k v)$$

with a fixed finite-dimensional vector space, operator T , vector v , and linear functional λ . Any such realization would give $\operatorname{rank} \mathbf{H}_w < \infty$.

This criterion is more flexible than a finite-state automaton. It rules out every exact finite-dimensional linear recurrence realization, including those over $\mathbb{R}$ or $\mathbb{C}$. It suggests a concrete operator target: derive finite Hankel rank, rationality of the generating function $\sum_{k \geq 0} w_k z^k$, or a uniform bounded-denominator recurrence from the simultaneous rationality of the two cocycles. Ordinary low-rank approximation is again insufficient: long finite Sturmian prefixes can be approximated by periodic words of growing periods.

14 Further operator routes: exact content and failure modes

This section records several natural operator-theoretic ideas that were tested against the local-to-global framework. The purpose is to prevent a reformulation from being mistaken for progress and to identify the small modification that would turn each route into a genuine closing argument.

14.1 Wold theory and unilateral multiplication shifts

On $\ell^2(\mathbb{N})$, the multiplication isometry $V_m e_n = e_{mn}$ is a pure isometry with countably infinite defect multiplicity for every $m \geq 2$. Hence individual operators V_m are all unitarily equivalent by Wold theory. A pair carries more orbit information, but its bilateral dilation returns to the universality of [Theorem 3.1](#). Therefore neither the spectrum of one V_m nor its Wold multiplicity can compare the labels $m = 2$ and $m = A$.

What would help: retain the additive shift S , the Hamiltonian H , or the concrete arithmetic basis. The affine relation with S is rigid by [Theorem 9.1](#); global H -covariance is rigid by [Theorem 6.1](#).

14.2 Tensor products and multiplicative closure

Tensor powers of semisimple scalar modes produce eigenvalues $2^a 3^b$. Unique factorization makes all such labels distinct, so the generated finite-dimensional functional calculus remains semisimple. Under the target map they become $A^a B^b$, which are again distinct. Therefore tensoring cannot spontaneously create the Jordan self-extension needed in [Theorem 11.1](#).

What would help: one additive decomposition that identifies two different tensor constructions, one nontrivial extension class, or one new simple object whose dimension is multiplicatively independent of 2, 3.

14.3 Cuntz algebras and K -theory

The Cuntz algebra $\mathcal{O}_n$ has [\[7\]](#) $K_0(\mathcal{O}_n) \cong \mathbb{Z}/(n-1)\mathbb{Z}$ and $K_1(\mathcal{O}_n) = 0$. Thus a stable isomorphism or Morita equivalence between $\mathcal{O}_2$ and $\mathcal{O}_A$ would force $A = 2$. But the scalar equality $A = 2^x$ supplies no such equivalence; postulating it already inserts a strong bridge between the source and target labels. At the other extreme, the K -theory of the fixed two-coordinate Toeplitz algebra does not see the weights $\log 2, \log 3$ at all.

What would help: a naturally constructed correspondence or bimodule whose K -theoretic class is forced by the scalar exponentiation and simultaneously relates $\mathcal{O}_2$ to $\mathcal{O}_A$. Without that construction, the K -groups merely restate that different known integers give different Cuntz algebras.

14.4 Operator monotonicity and Loewner positivity

For $0 < \alpha < 1$, the function t^α is operator monotone and operator concave on $(0, \infty)$. Its two-point Loewner matrix at 2, 3 is positive:

$$\begin{pmatrix} \alpha 2^{\alpha-1} & 3^\alpha - 2^\alpha \\ 3^\alpha - 2^\alpha & \alpha 3^{\alpha-1} \end{pmatrix} \succeq 0.$$

Under (1), the off-diagonal entry and the factors $2^{\alpha-1}, 3^{\alpha-1}$ are rational; the unknown α remains in the diagonal derivative. Positivity gives real inequalities, not an arithmetic discreteness condition.

What would help: algebraicity or rationality of any nonzero diagonal derivative, the determinant, or a finite-rank degeneracy. Each would make α algebraic and then force $\alpha = 0$, but none follows from positivity alone.

14.5 Spectral determinants and heat asymptotics

The local partition function (5), its meromorphic continuation, and the Weyl law (7) depend analytically on the continuous parameters $\log a, \log b$. Under $(a, b) \mapsto (a^x, b^x)$ they transform by ordinary scaling. Similarly, the Koopman cocycles have zero mean logarithm and bounded products by (16). Determinants and Lyapunov exponents therefore vanish or rescale exactly as they should for every real $x > 0$.

What would help: an external integral normalization of energy, such as the full prime spectrum, together with an exact finite-lattice implementation. Without the integral lattice, asymptotic coefficients are real-valued and cannot quantize the scale.

14.6 Continuous interpolation of degree

Maps $z \mapsto z^n$ on the circle, endomorphisms of $C(\mathbb{T})$, and finite-degree covering operators carry an integer topological degree. A norm-continuous family cannot change this degree. Consequently a continuous “fractional-degree” interpolation from 2 to A cannot remain inside the category of finite coverings.

This observation does not prove the conjecture: the hypotheses do not provide such an interpolation. It does show that attempts to realize $n \mapsto n^x$ as a continuous flow of genuine covering maps must leave the arithmetic category between integer times. The obstruction is topological rather than transcendence-theoretic.

14.7 Abstract dilation theory

The two-mode embedding J in (8) is an isometry, and abstract isometries admit many unitary or isometric dilations. Such dilations can assign arbitrary vectors to the orthogonal complement and therefore erase the valuation lattice. They do not imply that p^x is rational for a new prime p .

What would help: a dilation theorem inside a rigid category of monomial operators, product systems with integral degree maps, graph correspondences with finite incidence matrices, or based Hilbert modules whose basis is arithmetically distinguished. In any finite such category, Theorem 7.1 becomes the closing mechanism.

15 Five exact closing bridges

The operator analysis reduces a large search space to five compact targets. Each target adds information of a different kind, but each is provably sufficient.

15.1 Bridge I: one third algebraic mode

By [Theorem 2.1](#), prove $q^x \in \overline{\mathbb{Q}}$ for one $q \in \mathbb{Q}_{>0} \setminus \langle 2, 3 \rangle$. In Hamiltonian language, extend the partial intertwiner to one additional eigenvector:

$$Te_q \neq 0, \quad HT = xTH,$$

with Te_q supported at an algebraic arithmetic label. This is the smallest propagation target and uses the full strength of the Six Exponentials Theorem immediately.

15.2 Bridge II: one rational first jet

Construct any rational square-zero perturbation $N \neq 0$ for which $(2I + N)^x$ or $(3I + N)^x$ is rational. By [Theorem 11.1](#), this makes x rational. Conceptually, this requires a derivation or self-extension compatible with the scalar power operation.

A potentially useful algebraic formulation is to seek a $\mathbb{Q}$ -derivation δ on an operator algebra containing the scalar mode such that

$$\delta(a^x) = xa^{x-1}\delta(a)$$

holds and both $\delta(a)$ and $\delta(a^x)$ are rationally controlled. If $\delta(2) \neq 0$, the chain rule gives the missing coefficient.

15.3 Bridge III: finite prime-mode completion

Extend the two valuation columns

$$\nu(2) \mapsto \nu(A), \quad \nu(3) \mapsto \nu(B)$$

to an integral endomorphism of a finite prime lattice that scales the log-energy vector. Then [Theorem 7.1](#) applies. A unimodular extension is even more rigid and forces $x = 1$.

This target is natural for finite graph correspondences, substitution matrices, fusion rings, and finite product systems. The audit question for any proposed construction is simple: does its incidence matrix really satisfy $M^T \ell = x\ell$, or does it merely reproduce the two known columns?

15.4 Bridge IV: exact finite realization of the jump observable

Produce an exact finite-dimensional tracial model for P or an exact finite-dimensional linear realization of the Sturmian sequence (w_k) . The former invokes [Theorem 12.2](#); the latter invokes [Theorem 13.1](#). Approximation with growing dimensions or periods does not suffice.

A more operator-invariant variant uses the correlation sequence

$$c_k = \tau(PU_\alpha^k P U_\alpha^{-k}).$$

Its Fourier representation contains infinitely many nonzero frequencies when α is irrational. Any theorem forcing a finite spectral support, rational generating function, or bounded-denominator exact recurrence for (c_k) would likewise force rationality of α .

15.5 Bridge V: one additive–multiplicative coherence

Construct a morphism preserving one affine relation such as $V_2 S = S^2 V_2$, or one categorical relation such as $E_2 \otimes E_2 \cong E_2 \oplus E_2$. The robust gap in [Theorem 9.1](#) shows that even preservation modulo compact operators is enough in the concrete affine model.

A weaker propagation version would only need to manufacture a third mode. For example, a rule that makes the label $5 = 2 + 3$ available after exponentiation would give algebraicity of 5^x , and [Theorem 2.1](#) would finish the argument before full additive functoriality is required.

16 Prioritized research program

The following program is ordered by the amount of genuinely new arithmetic structure required, not by aesthetic appeal.

16.1 Priority 1: third-mode leakage from an additive transfer operator

The largest payoff per lemma is to derive algebraicity of 5^x (or any non-3-smooth rational base). The affine semigroup suggests studying transfer operators that mix translation and dilation:

$$(L_m f)(k) = \sum_{j=0}^{m-1} f\left(\frac{k+j}{m}\right)$$

in a suitable discrete or compact model, together with the multiplication isometries V_2, V_3 . The goal is not a full endomorphism of the $ax + b$ algebra—which would already force $x = 1$ —but a weaker positivity or integrality statement that produces one new algebraic mode.

A concrete test is whether the ranges or defect projections attached to V_2 and V_3 can be combined canonically to create the degree-five range projection. Any construction must be audited against [Theorem 3.1](#): if it uses only the commuting pair and not the additive placement of their ranges, it cannot distinguish the labels.

16.2 Priority 2: rational self-extension from a deformation

Seek a rational one-parameter deformation $T(t)$ through $T(0) = 2I$ such that $T(t)^x$ is rational to first order at $t = 0$. Differentiating would give

$$\left. \frac{d}{dt} T(t)^x \right|_{t=0} = x 2^{x-1} T'(0),$$

and any nonzero rational $T'(0)$ closes the problem.

Possible sources include:

- a derivation on a rational semigroup algebra;

- a nontrivial extension in a category generated by the two scalar modes and an arithmetic correspondence;
- a deformation of a transfer operator whose matrix entries remain rational because they count finite fibers;
- a divided-difference construction in which two spectral nodes are forced to coalesce without leaving the rational category.

The semisimple theorem shows exactly what must be avoided: coupling only the distinct nodes 2 and 3 gives $B - A$, never a derivative.

16.3 Priority 3: finite correspondence or fusion completion

Build a finite based module with basis indexed by a finite prime set and an integral endomorphism extending the two known valuation columns. The logarithmic dimension vector should be a left eigenvector with eigenvalue x . Once this is achieved, no delicate spectral estimate is needed; [Theorem 7.1](#) and Gelfond–Schneider finish the proof.

The hard point is not finding an integer matrix with the two desired columns; arbitrary completion is easy. The hard point is forcing all columns to obey the same logarithmic scaling. Candidate mechanisms are associativity constraints, duals in a rigid tensor category, finite index, or a trace/dimension function compatible with both source and target.

16.4 Priority 4: exact Hankel rigidity from two rational cocycles

The pair of rational-valued cocycles is stronger than one such cocycle only through the common exponent:

$$c_2 = 2^{P-\alpha I}, \quad c_3 = 3^{P-\alpha I}.$$

Both are affine rational functions of the same projection P , so purely algebraic elimination recovers P but not its trace. A successful argument must constrain the orbit of P , not merely P itself.

Promising exact statements would be:

1. a bounded-order recurrence for the words obtained from both cocycles;
2. a uniform denominator bound for all correlation moments;
3. an exact rational generating function for one nontrivial orbit observable;
4. a finite-dimensional representation whose dimension is bounded independently of the approximation scale.

Any of these would invoke [Theorem 13.1](#) or [Theorem 12.2](#).

16.5 Priority 5: quantitative affine defects

The gap $\|V_A S^d - S^{2d} V_A\|_{\text{ess}} \geq \sqrt{2}$ for $A \neq 2$ suggests looking for an arithmetic construction whose affine defect can be shown strictly smaller than $\sqrt{2}$. This is weaker than exact functoriality and robust under compact errors. Similar estimates can be studied in normalized

Schatten norms on finite truncations. In the standard basis, the wrong label produces a defect of fixed density, not a boundary effect.

The challenge is to connect the scalar equalities $A = 2^x, B = 3^x$ to such a norm estimate. Heat-kernel averaging or transfer-operator regularization might turn logarithmic scaling into approximate affine coherence, but any estimate must beat a fixed, representation-independent gap rather than merely tend to a nonzero constant.

17 Logical audit: what has and has not been proved

The conjecture itself remains open in this report. The rigorous gains are structural and conditional:

1. The largest rational/algebraic sector available under a counterexample is exactly the rank-two group generated by 2 and 3 ([Theorem 2.1](#)).
2. Pure bilateral shift data is completely universal ([Theorem 3.1](#)).
3. Local Hamiltonian, heat, and KMS data is exactly related by time rescaling ([Theorem 4.1](#)).
4. A global monomial Hamiltonian extension forces a positive integer exponent ([Theorem 6.1](#)).
5. A finite invariant prime lattice forces an integer exponent, and a lattice automorphism forces $x = 1$ ([Theorems 7.1](#) and [7.2](#)).
6. One affine relation is robustly label-rigid, even modulo compact operators ([Theorem 9.1](#)).
7. Semisimple rational functional calculus at 2, 3 is automatic and cannot help, while one Jordan self-extension forces rationality ([Theorems 10.1](#) and [11.1](#)).
8. The Koopman cocycles are bounded coboundaries; exact finite trace or Hankel realization closes, but approximation does not ([Theorems 12.1](#), [12.2](#) and [13.1](#)).

None of the additional structures in the right-hand column below has been derived from the scalar hypotheses.

Given unconditionally	Still missing
Two integer labels $A = 2^x, B = 3^x$	One third independent algebraic label q^x .
A rank-two covariant Fock embedding	A finite or global integral lattice completion.
All semisimple rational two-point lifts	One rational nonsemisimple self-extension.
Two rational-valued bounded coboundaries	An exact finite trace or finite-Hankel realization.
Pure multiplicative semigroup coherence	One additive–multiplicative coherence relation.

This distinction is essential. Each missing item is a genuine strengthening, not a disguised consequence already established in the report.

18 Updated literature reconnaissance

The literature search for this continuation was organized around three separate questions:

1. Has the special rational/integer case of the Four Exponentials problem, or the Alaoglu–Erdős conjecture itself, been resolved?
2. Have recent algebraic-independence results lowered the unconditional 2×3 Six Exponentials threshold to the needed 2×2 array in this arithmetic setting?
3. Do recent semigroup C^* -algebra, arithmetic-dynamics, or operator-algebra constructions supply a canonical additive or finite-lattice extension of the two multiplicative modes?

Searches were run through 21 August 2026 using the exact phrases “Alaoglu–Erdős conjecture”, “four exponentials conjecture”, “six exponentials theorem”, “Schanuel conjecture”, “algebraic independence logarithms”, “Bost–Connes”, “ $ax+b$ semigroup C^* -algebra”, and combinations with “arXiv”. Primary sources, arXiv records, journal pages, and author-hosted surveys were preferred.

The search found no verified unconditional resolution of the conjecture or of the Four Exponentials Conjecture. The operative unconditional threshold remains the Six Exponentials Theorem used in [Theorem 2.1](#). Modern work on Schanuel-type statements and exponential algebraic closedness [13, 23], period conjectures, explicit linear forms in logarithms, and zero estimates refines surrounding theory, but does not turn the particular vanishing determinant

$$(\log 2)(\log B) - (\log 3)(\log A) = 0$$

into a contradiction. Linear-form estimates apply when a nonzero linear form in logarithms must be bounded away from zero; here the hard relation is bilinear in logarithms and is exactly zero.

On the operator side, the Bost–Connes system, semigroup C^* -algebras of $ax + b$ type, product systems, and boundary quotients provide a mature language for the logarithmic Hamiltonian and for additive–multiplicative coherence. They support the distinction proved here: multiplicative time evolution is naturally homogeneous, whereas additive range relations remember integer degree. No source located in the search derives the required additive or finite-prime extension from the two scalar algebraicity assumptions alone.

The reproducible arXiv metadata ledger below records recent items returned by the focused queries. It is included to make the search auditable rather than to imply that every listed paper directly attacks the conjecture.

Date	Authors	Title	arXiv
<i>The live arXiv Atom harvest was not available inside the compilation sandbox. The search protocol and stable references in the bibliography remain reproducible; no unverified recent title has been inserted.</i>			

Remark 18.1 (How to interpret the ledger). The phrase searches inevitably return papers about model theory, exponential fields, linear forms, or operator systems that are adjacent rather than directly applicable. Their relevance is negative as well as positive: they confirm that the known unconditional theorem supplies three logarithmic columns, while the present conjecture supplies two. No claim of completeness is made for indexing services that were not openly searchable.

19 Conclusions

Operator theory does not bypass the transcendence barrier merely by placing the four algebraic numbers $2, 3, A, B$ on a Hilbert space. In the natural two-mode representations, it explains why the barrier is so stable: all multiplicative shift pairs are universal after bilateral amplification, and all local logarithmic Hamiltonians are related by a continuous time homothety. Spectra, heat traces, local equilibrium states, entropy, and bounded-coboundary invariants are therefore exactly as compatible with a nonintegral x as with an integral one.

The useful operator contribution is a precise map of where discreteness re-enters.

- The full arithmetic Hamiltonian has simple, discretely labelled spectrum. A global isometric rescaling must send every n to the integer n^x , and finite differences force $x \in \mathbb{N}$.
- A finite prime-mode completion turns logarithmic scaling into an integer-matrix eigenvalue problem. Gelfond–Schneider then forces the eigenvalue to be an integer.
- The affine relation $V_m S = S^m V_m$ remembers the integer label in a way the multiplicative pair cannot; even a compact perturbation cannot disguise a wrong label.
- Semisimple functional calculus sees only the two given values. A Jordan self-extension sees a derivative, and that first jet immediately rationalizes x .
- The Koopman model converts the fractional part into the trace of one projection. Exact finite-dimensional or finite-Hankel realization discretizes the trace; approximation does not.
- Six Exponentials shows that a hypothetical counterexample has a maximal algebraic sector of rank exactly two. One new independent algebraic mode is enough.

The most promising next attack is therefore not “find a subtler spectrum.” It is to derive one small coherence bridge that cannot be implemented in the universal rank-two model: a third mode, a first jet, a finite integral completion, an exact finite realization, or an additive range relation. These targets are narrow enough to test individual constructions decisively, yet strong enough that no further transcendence breakthrough would be needed once any one of them is obtained.

A Domain details for arithmetic Hamiltonians

This appendix records the unbounded-operator points used implicitly in the main text.

Let H be diagonal on $\ell^2(\mathbb{N})$, $H e_n = (\log n) e_n$, with maximal domain

$$\mathcal{D}(H) = \left\{ \xi = (\xi_n) : \sum_{n \geq 1} (\log n)^2 |\xi_n|^2 < \infty \right\}.$$

The finite-support subspace

$$\mathcal{C} = \text{span}\{e_n : n \in \mathbb{N}\}$$

is a core for H . If a bounded operator T satisfies $HT = cTH$ on $\mathcal{C}$, then $T\mathcal{C} \subseteq \mathcal{D}(H)$. For each n , spectral simplicity gives

$$T e_n \in \ker(H - c \log n).$$

Thus either $Te_n = 0$, or $n^c \in \mathbb{N}$ and $Te_n = \lambda_n e_{n^c}$ for some scalar λ_n . In particular:

Proposition A.1. *Every bounded H -covariant operator has the form*

$$Te_n = \begin{cases} \lambda_n e_{n^c}, & n^c \in \mathbb{N}, \\ 0, & n^c \notin \mathbb{N}, \end{cases}$$

with a bounded scalar family (λ_n) . It is an isometry exactly when $n^c \in \mathbb{N}$ and $|\lambda_n| = 1$ for every n .

This proposition explains why support, not spectral multiplicity, is the central invariant. Under a hypothetical counterexample, the support of every such algebraically labelled intertwiner is confined by [Theorem 2.1](#) to the 3-smooth sector.

On $\ell^2(\mathbb{Q}_{>0})$, the same argument applies to $H_G e_q = (\log q) e_q$. A covariant operator is monomial on the rational basis wherever $q^c \in \mathbb{Q}_{>0}$. Under a nonintegral Alaoglu–Erdős solution, [Theorem 2.1](#) says the algebraic support is precisely $\langle 2, 3 \rangle$.

B The finite-difference integrality lemma in greater generality

The proof of [Theorem 6.1](#) used a standard but useful criterion.

Theorem B.1 (Integer-valued real powers). *Let $c \geq 0$. If $n^c \in \mathbb{Z}$ for every sufficiently large integer n , then $c \in \mathbb{N}_0$.*

Proof. If $c \notin \mathbb{N}_0$, put $r = \lfloor c \rfloor + 1$. For all sufficiently large n , every term $(n+j)^c$, $0 \leq j \leq r$, is an integer, hence $\Delta^r n^c \in \mathbb{Z}$. The repeated-integral identity

$$\Delta^r f(n) = \int_{[0,1]^r} f^{(r)}(n + t_1 + \cdots + t_r) dt_1 \cdots dt_r$$

holds for $f \in C^r$. With $f(t) = t^c$, the derivative is

$$f^{(r)}(t) = c(c-1) \cdots (c-r+1) t^{c-r}.$$

It is nonzero with fixed sign on $(0, \infty)$, while $c-r < 0$. Therefore $\Delta^r n^c$ is nonzero and tends to zero, contradicting its eventual integrality. $\square$

The theorem remains valid if n^c is required to lie in any discrete subset of $\mathbb{R}$ with a uniform separation near the relevant range, provided the forward differences inherit that discreteness. This is one reason exact lattice-valued operator models are powerful, while asymptotic closeness to a lattice is not.

C Finite prime lattices as based operator correspondences

Let $P = \{p_1, \dots, p_d\}$ and let $\ell = (\log p_1, \dots, \log p_d)^\top$. An integer matrix M defines a monomial map on Laurent monomials:

$$z^v \mapsto z^{Mv}, \quad v \in \mathbb{Z}^d.$$

On the group Hilbert space $\ell^2(\mathbb{Z}^d)$, it defines $T_M e_v = e_{Mv}$ when M is injective. The energy relation is

$$H_\ell T_M e_v = \langle \ell, Mv \rangle e_{Mv} = \langle M^\top \ell, v \rangle e_{Mv}.$$

Hence $H_\ell T_M = c T_M H_\ell$ exactly when $M^\top \ell = c\ell$.

The same matrix can be viewed in several equivalent categories:

1. as an endomorphism of the free abelian multiplicative group Γ_P ;
2. as a monomial endomorphism of the algebraic torus $(\mathbb{C}^\times)^d$;
3. as a based correspondence on the valuation lattice;
4. when $M \geq 0$, as a multi-shift on the positive Fock cone;
5. as an incidence matrix with logarithmic dimension vector ℓ .

In every finite realization, the scaling factor is an eigenvalue of a monic integer polynomial. This is the categorical content of [Theorem 7.1](#). Passing to an infinite prime set removes the algebraic-integer conclusion: an infinite integer matrix may have arbitrary positive eigenvalues in an appropriate functional-analytic sense. Therefore finiteness is not cosmetic; it is the source of arithmetic discreteness.

C.1 A completion test

Suppose a construction proposes a finite matrix M extending the two known columns. The following checklist detects circularity.

1. Are the columns genuinely integral valuation vectors?
2. Does $M^\top \ell = x\ell$ hold for *every* column, or only for the columns indexed by 2 and 3?
3. Is the prime set finite and invariant under all introduced power images?
4. Was rationality of a new p^x proved, or silently assumed when its valuation column was written down?
5. If M is claimed unimodular, is surjectivity of the power map actually available?

A positive answer to the first three questions, without assuming the desired conclusion in the fourth, is already a proof.

D Functional calculus on primary components

Let T be a rational matrix with positive spectrum contained in $\{2, 3\}$. Its minimal polynomial has the form

$$m_T(t) = (t - 2)^r (t - 3)^s.$$

Hermite interpolation gives a polynomial $P_x(t)$ of degree less than $r + s$ such that

$$P_x^{(j)}(2) = \left. \frac{d^j}{dt^j} t^x \right|_{t=2}, \quad 0 \leq j < r,$$

and similarly at 3. Then $T^x = P_x(T)$. Consequently:

Proposition D.1 (Jet hierarchy). *Under $2^x, 3^x \in \mathbb{Q}$, rationality of T^x is automatic for every semisimple T ($r, s \leq 1$). If $r \geq 2$ or $s \geq 2$, it additionally requires rational control of at least one derivative*

$$(x)_j 2^{x-j} \quad \text{or} \quad (x)_j 3^{x-j},$$

where $(x)_j = x(x-1)\cdots(x-j+1)$. The first nonzero self-extension, $j=1$, already forces $x \in \mathbb{Q}$.

This hierarchy precisely classifies the information visible to every finite-dimensional rational functional-calculus model supported at the two original bases.

D.1 Why the distinct-node triangular matrix is split

For completeness, the matrix

$$T_c = \begin{pmatrix} 2 & c \\ 0 & 3 \end{pmatrix}$$

is rationally diagonalizable:

$$\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}^{-1} T_c \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

after the appropriate sign convention for the off-diagonal conjugating entry. More invariantly, the coprimality of $t-2$ and $t-3$ makes the module split. Its off-diagonal entry in T_c^x is a divided difference, not a derivative. A genuine Jordan block cannot be reached by a rational similarity while keeping distinct eigenvalues.

E Koopman calculations and correlation data

The jump identity

$$\sum_{j=0}^{N-1} (w(R_\alpha^j t) - \alpha) = t - R_\alpha^N t$$

proves bounded discrepancy directly. In particular, the Sturmian word has balanced prefixes:

$$\left| \sum_{j=0}^{N-1} w_j - N\alpha \right| < 1.$$

This strong regularity does not imply periodicity; it is shared by every irrational rotation word.

The distinguished interval projection has Fourier coefficients

$$\hat{w}(k) = \begin{cases} \alpha, & k = 0, \\ \frac{e^{-2\pi i k(1-\alpha)} - e^{-2\pi i k}}{2\pi i k}, & k \neq 0. \end{cases}$$

For irrational α , infinitely many coefficients are nonzero. The correlation sequence

$$c_n = \tau(PU_\alpha^n P U_\alpha^{-n})$$

has the Fourier expansion

$$c_n = \sum_{k \in \mathbb{Z}} |\widehat{w}(k)|^2 e^{2\pi i k n \alpha}.$$

Thus its spectral measure has infinitely many atoms for irrational α . Any exact representation that forces only finitely many frequencies would make α rational. This is the correlation counterpart of the Hankel-rank criterion.

E.1 Exact versus asymptotic matrix models

There are three materially different claims:

1. *Exact fixed-size model*: a single M_N reproduces the projection and its trace exactly. This forces $\alpha \in \frac{1}{N}\mathbb{Z}$.
2. *Uniformly bounded size*: a sequence of exact models has dimensions bounded by N_0 . Passing to a constant-size subsequence again forces a rational trace with bounded denominator.
3. *Growing-size approximation*: dimensions tend to infinity and traces converge to α . This is compatible with every real α and gives no arithmetic conclusion.

Any future claim based on microstates or finite sections should specify which of these regimes is proved.

F Dependency map for the closing criteria

The logical flow can be summarized without operator terminology.

third independent algebraic mode $\implies x \in \mathbb{Q}$	(Six Exponentials),
rational first derivative at 2 or 3 $\implies x \in \mathbb{Q}$	(elementary division),
finite invariant prime lattice $\implies x$ algebraic integer	(integer matrix),
x algebraic and $2^x \in \overline{\mathbb{Q}} \implies x \in \mathbb{Q}$	(Gelfond–Schneider),
$x \in \mathbb{Q}, 2^x \in \mathbb{Q} \implies x \in \mathbb{Z}$	(valuation),
exact finite trace/Hankel model $\implies \alpha \in \mathbb{Q}$	(finite rank/periodicity),
$\alpha \in \mathbb{Q}, 2^\alpha \in \mathbb{Q} \implies \alpha = 0$	(valuation),
one affine coherence $\implies A = 2$	(translation length),
$A = 2^x = 2 \implies x = 1$.	

This map makes the role of each external theorem explicit and prevents a closing criterion from being confused with a derivation of its premise.

G Reproducible search notes

The arXiv metadata queries used for the ledger were variants of:


```
all:"four exponentials conjecture"
(all:Schanuel OR all:"six exponentials" OR
 all:"four exponentials") AND cat:math.NT
(all:"algebraic independence" AND all:logarithms) AND cat:math.NT
(all:"Bost-Connes" OR all:"ax+b semigroup" OR
 all:"semigroup  $C^*$ -algebra") AND cat:math.OA
```

Results were sorted by submission date and screened for direct relevance. General web searches additionally used the exact conjecture name with the scalar condition $2^x, 3^x \in \mathbb{Z}$, and checked cited surveys and journal versions. Because negative literature searches can never prove nonexistence, the report's mathematical claims rely only on the theorems proved here or on standard cited results, not on the absence of a search hit.

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